

PRACTICA N°4

PRUEBAS DE RENDIMIENTO CON APACHE JMeter

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Materia: Verificación y Validación de Software – INF 780

Fecha: 01/07/26

GitHub: <https://github.com/nigthmare1230-hash/-INF780-Practica4-JMeter.git>

Parte 1 — Instalación y verificación

1. Descargue Apache JMeter 5.6.x desde jmeter.apache.org (requiere Java 8 o superior).
2. Descomprima y ejecute bin/jmeter (Windows: jmeter.bat; Linux/Mac: ./jmeter).
3. Confirme que la API está levantada y respondiendo en localhost:3000.
4. Capture la pantalla de inicio de JMeter y la versión de Java (java -version).

```
C:\Users\Usuario\Nueva carpeta (2)\INF780-MoviesBackend>java -version
java version "1.8.0_471"
Java(TM) SE Runtime Environment (build 1.8.0_471-b09)
Java HotSpot(TM) 64-Bit Server VM (build 25.471-b09, mixed mode)

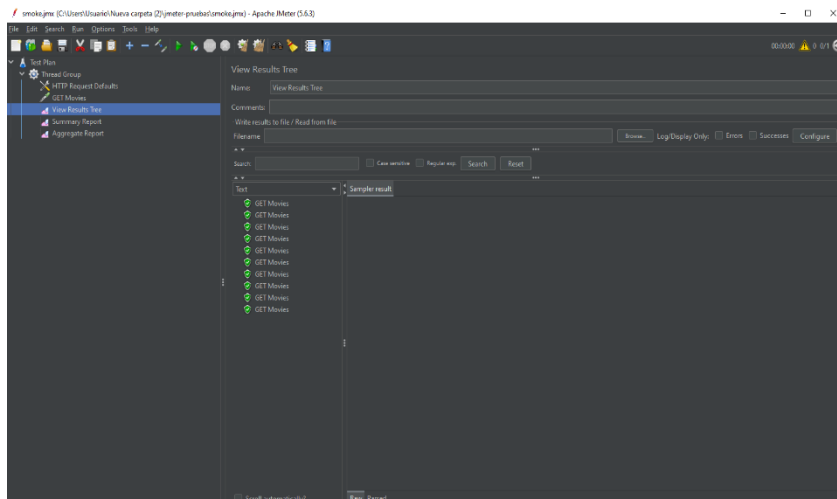
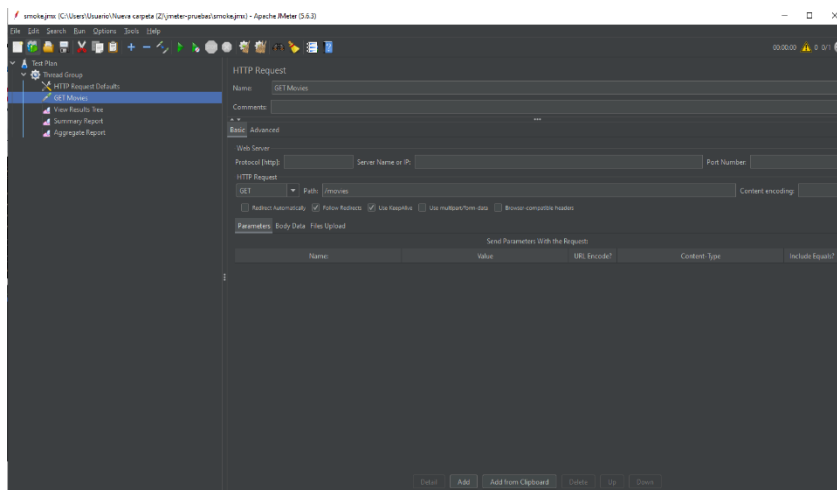
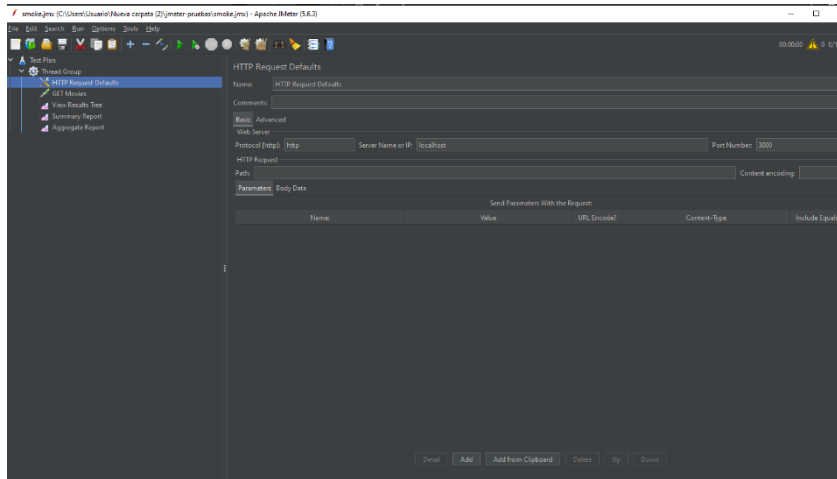
C:\Users\Usuario\Nueva carpeta (2)\INF780-MoviesBackend>jmeter --version
WARN StatusConsoleListener The use of package scanning to locate plugins is deprecated and will be removed in a future
release
WARN StatusConsoleListener The use of package scanning to locate plugins is deprecated and will be removed in a future
release
WARN StatusConsoleListener The use of package scanning to locate plugins is deprecated and will be removed in a future
release
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release

  APACHE JMeter 5.6.3

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```

Parte 2 — Plan base (smoke test)

1. Cree un Test Plan y agregue un HTTP Request Defaults (Server Name: localhost, Port: 3000).
2. Agregue un Thread Group: 1 hilo, ramp-up 1 s, loops 5.
3. Agregue un HTTP Request GET /movies.
4. Agregue los listeners: View Results Tree, Summary Report y Aggregate Report.
5. Ejecute y verifique que todas las peticiones son exitosas (Error % = 0). Guarde como smoke.jmx.



smoke.jmx (C:\Users\Usuario\Nueva carpeta (2)\meter-pruebas\smoke.jmx) - Apache JMeter (3.6.3)

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors: ☐ Successes: ☐ Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/s	Sent KB/s	Avg. Bytes
GET Movies	10	24	2	185	34.43	0.00%	12.8/sec	13.23	0.03	63182.0
TOTAL	10	24	2	185	34.43	0.00%	12.8/sec	13.23	0.03	63182.0

☐ Include group name in label? ☒ Save Table Header

smoke.jmx (C:\Users\Usuario\Nueva carpeta (2)\meter-pruebas\smoke.jmx) - Apache JMeter (3.6.3)

Aggregate Report

Name: Aggregate Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors: ☐ Successes: ☐ Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB/s	Sent KB/s
GET Movies	10	24	4	22	32	185	2	185	0.00%	12.8/sec	13.23	0.03
TOTAL	10	24	4	22	32	185	2	185	0.00%	12.8/sec	13.23	0.03

☐ Include group name in label? ☒ Save Table Header

Parte 3 — Prueba de carga

1. Duplique el plan. Configure el Thread Group a 50 usuarios, ramp-up 30 s, loops 10.
2. Agregue al menos 3 peticiones distintas: GET /movies, GET /movies/{id} (con CSV Data Set Config de IDs) y POST /movies.
3. Agregue un Constant Timer o Gaussian Random Timer (p. ej. 300 ms) para simular usuarios reales.
4. Ejecute la prueba y registre throughput, media, percentil 90/95 y Error % de cada petición.
5. Guarde como carga.jmx y exporte los resultados a carga.jtl.

carga.jmx (C:\Users\Usuario\Nueva carpeta (2)\meter-pruebas\carga.jmx) - Apache JMeter (3.6.3)

Thread Group

Name: Thread Group

Comments:

Action to be taken after a Sampler error

☒ Continue ☐ Start Next Thread Loop ☐ Stop Thread ☐ Stop Test ☐ Stop Test Now

Thread Properties

Number of Threads (users): 50

Ramp-up period (seconds): 30

Loop Count: ☐ Infinite 10

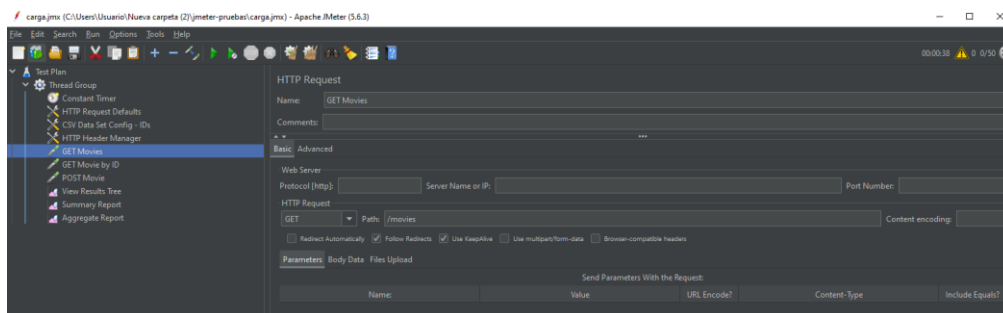
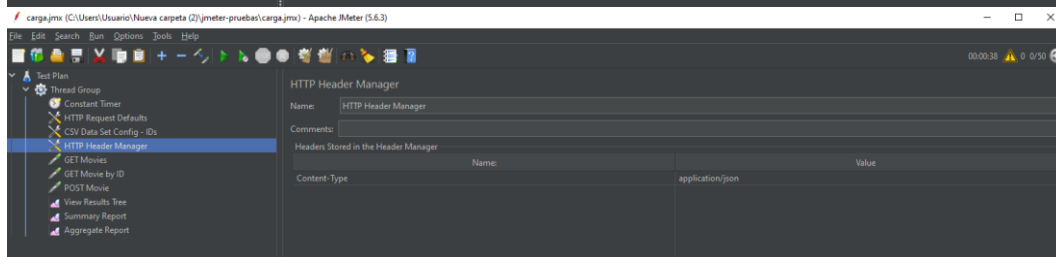
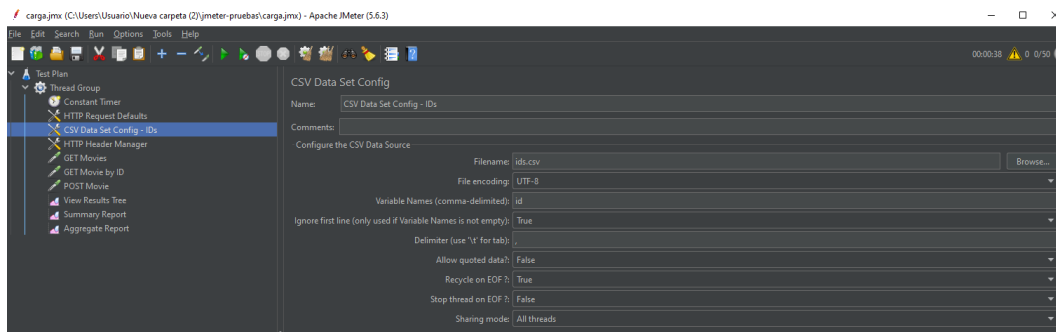
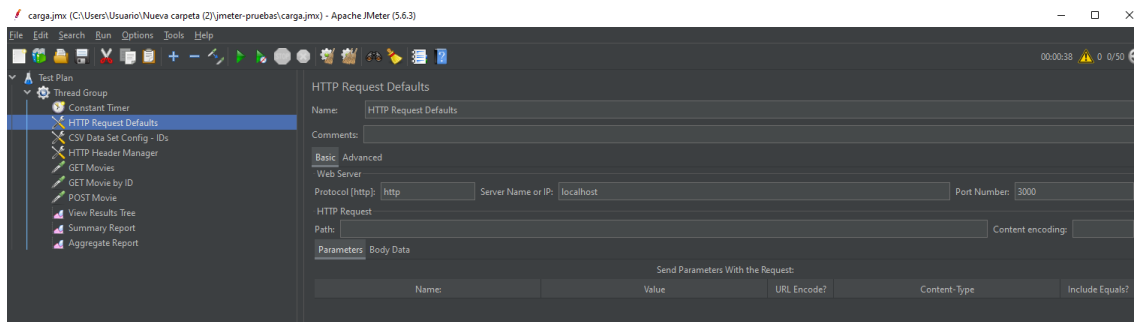
☒ Same user on each iteration

☐ Delay Thread creation until needed

☐ Specify Thread Lifetime

Duration (seconds):

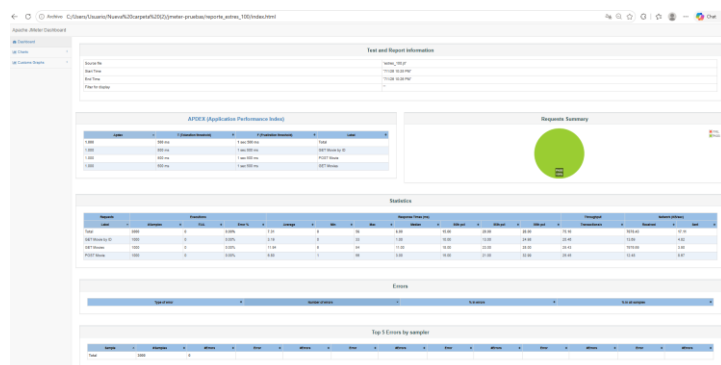
Startup delay (seconds):

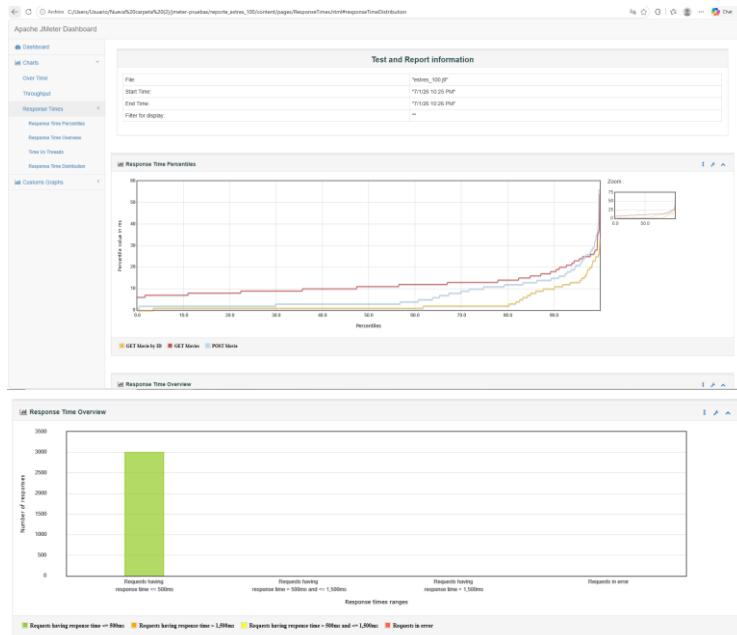


Parte 4 — Prueba de estrés

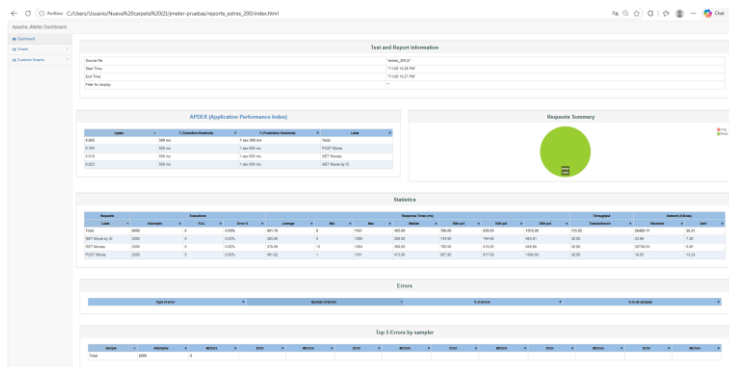
1. Reutilice el plan de carga aumentando los usuarios en etapas: 100, 200 y 400 (puede usar varias corridas o el plugin Ultimate Thread Group).
2. Registre en qué nivel comienzan a aparecer errores o los tiempos de respuesta se disparan.
3. Identifique el punto de saturación (máxima carga con Error % aceptable, p. ej. < 1%).
4. Guarde el plan como `estres.jmx` y documente la tabla de resultados por nivel.

Etapas 1

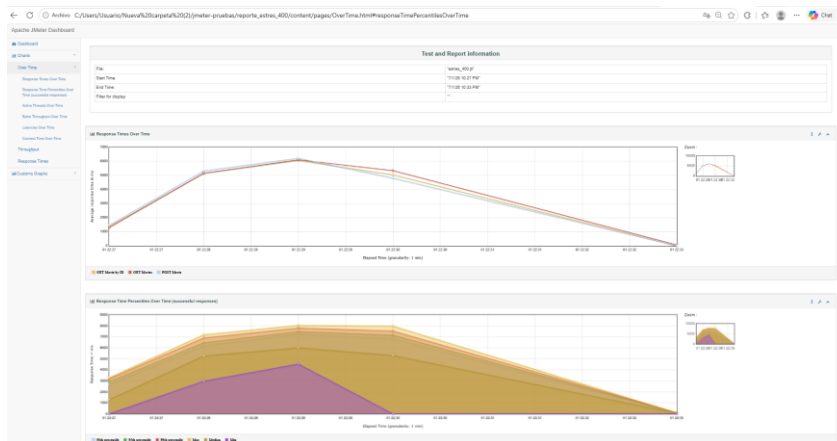
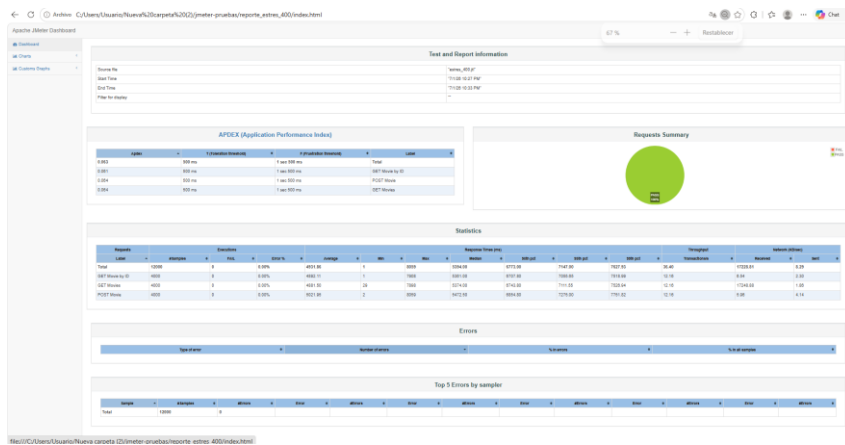




Etapa 2



Etapa 3



Parte 5 — Prueba de picos y aserciones

1. Configure un escenario de pico: arranque brusco (ramp-up corto, p. ej. 200 usuarios en 5 s) y observe la recuperación.
2. Agregue una Response Assertion que valide código de respuesta 200 (o 201 en POST).
3. Agregue una Duration Assertion con un umbral acordado (p. ej. 800 ms) y observe cuántas peticiones lo incumplen bajo carga.
4. Guarde como picos.jmx.

